Discourse Processing in NLP

Unit 4 - NLP

Introduction to Discourse Processing

What is Discourse?

- Discourse refers to a **connected sequence of sentences** that together convey a complete meaning.
- It goes beyond individual sentences and focuses on how sentences are **logically and semantically linked**.
- It is often described as **"language above the sentence level"**.

Key Characteristics:

- Sentences are not independent; they depend on each other.
- Meaning is derived using **context and background knowledge**.
- Includes understanding of **speaker intention and real-world knowledge**.

Introduction to Discourse Processing(Contd)

Example:

Ravi dropped the glass. It broke.

Explanation:

- The word “It” refers to “glass”
- The second sentence is logically connected (cause-effect)
- This connection is not explicitly stated but inferred

Why Discourse Processing is Important

Limitations of Sentence-Level Analysis:

- Sentence-level NLP cannot capture:
  - Pronoun references
  - Hidden meanings
  - Relationships between sentences

Role of Discourse Processing:

- Resolves **ambiguity**
- Identifies **relationships between sentences**
- Helps in understanding **implicit meaning and intention**
- Uses **world knowledge** and **context**

Example:

> Excuse me, you are standing on my foot.

Interpretation:

- Literal meaning: Statement
- Actual meaning: **Polite request to move**

Discourse vs Sentence-Level Processing

Sentence-Level Processing:

- Focuses on grammar and syntax
- Each sentence treated independently
- Example: Parsing, POS tagging

Discourse-Level Processing:

- Focuses on meaning across sentences
- Considers context and relationships
- Handles:
  - Reference resolution
  - Coherence
  - Speaker intention

Illustration:

John went to the bank. He deposited money.

- "He" refers to John
- "bank" interpreted as financial institution (context-based)

Knowledge Required in Discourse Processing

Types of Knowledge Used:

- **Linguistic Knowledge** – grammar, syntax
- **Contextual Knowledge** – surrounding text
- **World Knowledge** – real-world facts
- **Situational Knowledge** – current situation
- **Cultural Knowledge** – social norms

Example:

The trophy would not fit in the suitcase because it was too big.

Explanation:

- "it" refers to trophy (requires reasoning)
- Requires world knowledge about size relationships

Student Engagement Questions (Introduction)

Conceptual Questions:

- What is discourse and how is it different from a sentence?
- Why is context important in understanding language?
- Can a grammatically correct text be meaningless? Give example.

Analytical Questions:

- Identify the reference in: "Ravi lost his keys. He is upset."
- Explain how intention changes meaning in communication.

Application-Based:

- How is discourse processing useful in chatbots?
- Why is it important in machine translation?

Cohesion in Discourse Processing (Detailed)

Origin of Cohesion:

- The concept of cohesion was introduced by **Halliday and Hasan (1976)**
- Presented in the book: *"Cohesion in English"*
- They defined cohesion as the **linguistic glue** that binds sentences together

What is Cohesion?

- Cohesion refers to the **surface-level grammatical and lexical connections** between sentences
- It ensures that a text is **connected, readable, and interpretable**
- Achieved using explicit linguistic devices such as:
  - Pronouns (he, she, it)
  - Repetition of words
  - Synonyms
  - Conjunctions

Cohesion in Discourse Processing (Detailed)

Key Idea:

- Cohesion answers: **How are sentences linked together?**
- It does NOT guarantee meaning (that is coherence)

Why Cohesion is Needed

Without Cohesion:

Ravi bought a laptop. The weather is nice. Cats are sleeping.

- Grammatically correct but disconnected

With Cohesion:

Ravi bought a laptop. It is very fast and useful for his work.

Why Cohesion is Important:

- Ensures **text continuity**
- Reduces unnecessary repetition
- Improves **readability and clarity**
- Helps in **reference resolution**
- Enables machines to process multi-sentence meaning

In NLP Systems:

- Chatbots must track references
- Machine translation must preserve cohesion
- Summarization must maintain flow

Types of Cohesion

1. **Grammatical Cohesion**
   - Reference (pronouns)
   - Ellipsis (omission)
   - Substitution
   - Conjunction
2. **Lexical Cohesion**
   - Repetition
   - Synonym
   - Hypernym (generalization)

Key Insight:

- Grammatical cohesion uses **structure**
- Lexical cohesion uses **meaning relationships**

Examples with Explanation

Example 1: Reference

Ravi bought a car. It is expensive.

- "It" refers to "car"
- Avoids repetition → cohesive

Example 2: Repetition

The dog barked loudly. The dog was angry.

- Repetition of "dog" maintains cohesion

Example 3: Synonym

The boy is intelligent. The child solves problems quickly.

- "boy" and "child" are semantically related

Worked Example (Step-by-Step)

Text:

Suha bought a printer. The printer was expensive. It stopped working.

Step 1: Identify Cohesive Devices

- "printer" repeated → lexical cohesion
- "It" → reference

Step 2: Resolve Reference

- "It" refers to "printer"

Step 3: Analyze Flow

- Sentence 1 → introduces entity
- Sentence 2 → adds detail
- Sentence 3 → continues same entity

Conclusion:

- Strong cohesion through repetition + pronoun

Applications of Cohesion in NLP

1. **Machine Translation**
   - Ensures pronouns are correctly translated
2. **Chatbots**
   - Maintain conversation context
3. **Text Summarization**
   - Produces coherent summaries
4. **Information Extraction**
   - Links entities across sentences
5. **Question Answering**
   - Resolves references to answer correctly

Student Practice Problems

Identify Cohesion Type:

- Ravi has a dog. It is very friendly.
- The teacher entered the room. The instructor was angry.

Find Reference:

- Sita bought a book. She read it.

Spot Errors:

- Ravi bought a laptop. She is working well.

Improve Cohesion:

Ravi bought a phone. Ravi uses the phone daily.

Rewrite:

- Replace repetition with pronoun

Advanced Conceptual Questions

Theory Questions:

- Define cohesion with examples
- Explain types of cohesion
- Difference between cohesion and coherence

Analytical Questions:

- Analyze cohesion in a paragraph
- Identify all cohesive devices in given text

Exam-Oriented (10 Marks):

- Explain cohesion with types and examples
- Discuss role of cohesion in NLP applications

Reference in Discourse Processing (Detailed)

Origin of Reference:

- Introduced by **Halliday and Hasan (1976)**
- Part of **grammatical cohesion**
- Helps connect expressions to entities in discourse

What is Reference?

- Reference is the process of **linking linguistic expressions to real-world or textual entities**
- It helps identify **who or what is being talked about**
- Essential for understanding multi-sentence meaning

Key Idea:

- Reference answers: **"What does this word point to?"**
- Common in pronouns, articles, and demonstratives

Why Reference is Needed

Without Reference:

Ravi bought a laptop. Ravi uses the laptop daily.

With Reference:

Ravi bought a laptop. He uses it daily.

Importance:

- Avoids repetition
- Improves readability
- Maintains discourse flow
- Enables understanding of connected text

In NLP Systems:

- Chatbots track users and objects
- Machine translation resolves pronouns
- Question answering systems identify entities

Types of Reference

1. **Indefinite Reference**
   - Introduces a new entity
   - Example: *a book, an apple*
2. **Definite Reference**
   - Refers to known entity
   - Example: *the book*
3. **Pronominal Reference**
   - Uses pronouns
   - Example: *he, she, it*
4. **Demonstrative Reference**
   - Indicates location or proximity
   - Example: *this, that*
5. **Ordinal Reference**
   - Uses ordering
   - Example: *first, second*

Anaphora and Cataphora

Anaphora (Backward Reference):

Ravi bought a car. It is expensive.

- "It" refers to earlier noun (car)

Cataphora (Forward Reference):

When he arrived, Ravi was tired.

- "He" refers to Ravi (appears later)

Insight:

- Anaphora is more common in discourse

Examples with Explanation

Example 1: Pronoun Reference

Sita bought a book. She read it.

- "She" $\rightarrow$ Sita
- "it" $\rightarrow$ book

Example 2: Definite Reference

I saw a dog. The dog was barking.

- "The dog" refers to previously mentioned dog

Example 3: Demonstrative

I bought a phone. This is very useful.

- "This" refers to phone

Worked Example (Step-by-Step)

Text:

Ravi met Mohan. He greeted him warmly.

Step 1: Identify Pronouns

- "He", "him"

Step 2: Possible Candidates

- Ravi
- Mohan

Step 3: Apply Reasoning

- Subject preference $\rightarrow$ Ravi likely "He"
- "him" $\rightarrow$ Mohan

Final Interpretation:

- Ravi greeted Mohan

Applications of Reference

1. **Chatbots**
   - Maintain conversation context
2. **Machine Translation**
   - Correct pronoun mapping across languages
3. **Question Answering**
   - Identifies entities correctly
4. **Text Summarization**
   - Maintains reference consistency
5. **Information Extraction**
   - Links mentions of same entity

Student Practice Problems

Identify Reference:

- Ravi bought a car. It is fast.
- Sita met Geeta. She smiled.

Find Ambiguity:

- Ravi met Mohan. He was tired.

Correct the Sentence:

- Ravi bought a laptop. She is working well.

Rewrite for Better Reference:

Ravi bought a phone. Ravi uses the phone daily.

Advanced Questions (Exam-Oriented)

Theory Questions:

- Define reference with examples
- Explain types of reference
- Difference between anaphora and cataphora

Analytical Questions:

- Resolve references in given paragraph
- Identify ambiguous references

10-Mark Questions:

- Explain reference and its types with examples
- Discuss importance of reference in NLP

Ellipsis in Discourse Processing (Detailed)

Origin of Ellipsis:

- Introduced in **Halliday and Hasan (1976)** under grammatical cohesion
- Considered a mechanism for **economy of language**
- Widely used in both spoken and written discourse

What is Ellipsis?

- Ellipsis is the **omission of words or phrases** that are understood from context
- The missing information is **recoverable** from previous text
- Avoids redundancy while maintaining meaning

Key Idea:

- Ellipsis answers: **"What is missing but understood?"**

Why Ellipsis is Needed

Without Ellipsis:

Do you like fish?
Yes, I like fish.

With Ellipsis:

Do you like fish?
Yes, I do.

Importance:

- Reduces repetition
- Makes communication natural and efficient
- Improves fluency in conversation
- Maintains cohesion without redundancy

In NLP Systems:

- Dialogue systems must infer missing words
- Chatbots must reconstruct full meaning
- Machine translation must restore omitted content

Types of Ellipsis

1. **Nominal Ellipsis**
   - Omission of noun
   - Example: I like the red shirt, not the blue *(shirt)*
2. **Verbal Ellipsis**
   - Omission of verb phrase
   - Example: She can play the guitar, and he can *(play the guitar)* too
3. **Clausal Ellipsis**
   - Omission of clause
   - Example: Will you come? Yes *(I will come)*

Insight:

- Type depends on what part of sentence is omitted

Examples with Explanation

Example 1: Verbal Ellipsis

Do you like coffee?
Yes, I do.

- "like coffee" is omitted

Example 2: Nominal Ellipsis

I prefer the red shirt, not the blue.

- "shirt" is omitted after "blue"

Example 3: Clausal Ellipsis

Who finished the work?
Ravi.

- Full sentence omitted: "Ravi finished the work"

Worked Example (Step-by-Step)

Text:

Ravi can solve the problem. Mohan can too.

Step 1: Identify Ellipsis

- “can too” is incomplete

Step 2: Recover Missing Content

- Missing phrase: “solve the problem”

Step 3: Reconstructed Sentence

- Mohan can solve the problem too

Conclusion:

- Ellipsis reduces repetition but meaning remains clear

Applications of Ellipsis

1. **Dialogue Systems**
   - Understand incomplete responses
2. **Chatbots**
   - Handle short replies like "Yes", "No", "Maybe"
3. **Machine Translation**
   - Restore omitted words in target language
4. **Speech Recognition**
   - Interpret incomplete spoken sentences
5. **Text Summarization**
   - Remove redundant information

Student Practice Problems

Identify Ellipsis:

- She likes apples, and he does too.
- I ordered tea, not coffee.

Recover Missing Words:

- Ravi can sing, and Sita can too.

Find Type of Ellipsis:

- Will you attend the meeting? Yes.

Rewrite Fully:

Do you play cricket? Yes, I do.

Advanced Questions (Exam-Oriented)

Theory Questions:

- Define ellipsis with examples
- Explain types of ellipsis

Analytical Questions:

- Identify ellipsis in paragraph
- Recover omitted words

10-Mark Questions:

- Explain ellipsis with types and examples
- Discuss role of ellipsis in discourse processing

Lexical Cohesion in Discourse Processing (Detailed)

Origin of Lexical Cohesion:

- Introduced by **Halliday and Hasan (1976)**
- Part of **cohesion theory** in discourse analysis
- Focuses on **meaning-based connections** rather than grammar

What is Lexical Cohesion?

- Lexical cohesion refers to **connections between words based on meaning relationships**
- It uses vocabulary choices to link sentences
- Helps maintain **semantic continuity** in text

Key Idea:

- Lexical cohesion answers: **"How are words semantically related?"**

Difference: Cohesion vs Lexical Cohesion

Cohesion:

- Broad concept of **linking sentences in a text**
- Includes both **grammatical and lexical connections**
- Uses:
  - Reference (he, she, it)
  - Ellipsis (omission)
  - Substitution
  - Conjunctions
- Focuses on **surface-level structure**

Lexical Cohesion:

- Subtype of cohesion based on **word meaning relationships**
- Uses vocabulary-based connections
- Includes:
  - Repetition
  - Synonyms
  - Hypernyms/Hyponyms
  - Antonyms
- Focuses on **semantic (meaning-based) connections**

Difference: Cohesion vs Lexical Cohesion

Example:

Ravi bought a car. He loves the vehicle.

- "He" $\rightarrow$ Cohesion (reference)
- "car $\rightarrow$ vehicle" $\rightarrow$ Lexical cohesion

Key Insight:

- **All lexical cohesion is cohesion, but not all cohesion is lexical**

Why Lexical Cohesion is Needed

Without Lexical Cohesion:

Ravi bought a car. The weather is nice. Apples are tasty.

- No semantic connection between sentences

With Lexical Cohesion:

Ravi bought a car. The vehicle is very fast and comfortable.

Importance:

- Maintains **topic continuity**
- Improves **text readability**
- Helps in **semantic interpretation**
- Supports discourse coherence

In NLP Systems:

- Identifies related terms
- Improves summarization and search
- Helps in topic modeling

Types of Lexical Cohesion

1. **Repetition**
   - Same word repeated
   - Example: *The dog barked. The dog was angry.*
2. **Synonym**
   - Different words with similar meaning
   - Example: *boy* → *child*
3. **Antonym**
   - Opposite meaning
   - Example: *hot* → *cold*
4. **Hypernym (Generalization)**
   - General category word
   - Example: *car* → *vehicle*
5. **Hyponym (Specialization)**
   - Specific instance
   - Example: *vehicle* → *car*

Examples with Explanation

Example 1: Repetition

The cat is sleeping. The cat looks tired.

- Repetition of "cat" maintains cohesion

Example 2: Synonym

The boy is playing. The child is happy.

- "boy" and "child" are semantically related

Example 3: Hypernym

Ravi bought a car. The vehicle is expensive.

- "vehicle" is a general term for "car"

Worked Example (Step-by-Step)

Text:

The lion is strong. The animal lives in the forest. This predator hunts deer.

Step 1: Identify Words

- lion, animal, predator

Step 2: Identify Relationships

- lion $\rightarrow$ animal (hypernym)
- lion $\rightarrow$ predator (semantic relation)

Step 3: Analyze Cohesion

- All words refer to same entity
- Maintains topic continuity

Conclusion:

- Strong lexical cohesion through semantic relations

Applications of Lexical Cohesion

1. **Text Summarization**
   - Identifies key terms and topics
2. **Information Retrieval**
   - Matches related words (query expansion)
3. **Machine Translation**
   - Preserves meaning relationships
4. **Topic Modeling**
   - Detects themes in text
5. **Semantic Search**
   - Finds related concepts, not just exact words

Student Practice Problems

Identify Type:

- The dog barked. The animal ran away.
- The boy is happy. The child is smiling.

Find Lexical Relation:

- Car $\rightarrow$ vehicle
- Rose $\rightarrow$ flower

Improve Cohesion:

Ravi bought a car. Ravi drives the car daily.

Rewrite using lexical cohesion

Spot Error:

- Ravi bought a car. The apple is red.

Advanced Questions (Exam-Oriented)

Theory Questions:

- Define lexical cohesion with examples
- Explain types of lexical cohesion

Analytical Questions:

- Identify lexical cohesion in paragraph
- Explain semantic relations in text

10-Mark Questions:

- Explain lexical cohesion with types and examples
- Discuss role of lexical cohesion in NLP

Reference Resolution in Discourse Processing (Detailed)

Origin of Reference Resolution:

- Rooted in **AI and computational linguistics**
- Studied under **anaphora resolution**
- Important in discourse processing and NLP systems

What is Reference Resolution?

- Process of **identifying the entity referred to by a pronoun or expression**
- Links pronouns to their **correct antecedents**
- Essential for understanding multi-sentence meaning

Key Idea:

- Answers: **"Who or what does this refer to?"**

Why Reference Resolution is Needed

Without Resolution:

Ravi met Sita. She smiled.

- "She" is ambiguous without interpretation

With Resolution:

- "She" $\rightarrow$ Sita

Importance:

- Resolves ambiguity
- Maintains discourse continuity
- Enables correct interpretation of text

In NLP Systems:

- Chatbots track entities in conversation
- Machine translation resolves pronouns correctly
- QA systems identify correct answers

Constraints in Reference Resolution

1. **Person Agreement**
   - Pronoun must match person (1st, 2nd, 3rd)
2. **Gender Agreement**
   - He → male, She → female
3. **Number Agreement**
   - Singular vs plural

Example:

Ravi bought a laptop. She is working well.

Explanation:

- "She" does not match "laptop" → invalid reference

Preferences in Reference Resolution

1. **Recency Preference**
   - Most recent entity is preferred
2. **Subject Preference**
   - Subject is more likely than object
3. **Parallelism**
   - Similar grammatical roles preferred

Example:

Ravi met Mohan. He smiled.

- "He" $\rightarrow$ Ravi (subject preference)

Pronoun Resolution Algorithm

Steps:

1. Identify all possible candidates
2. Apply constraints (gender, number, person)
3. Apply preferences (recency, subject)
4. Assign salience score
5. Select best candidate

Key Concept:

- **Salience** = importance score of entity

Worked Example (Step-by-Step)

Text:

Ravi met Mohan. He greeted him.

Step 1: Identify Pronouns

- He, him

Step 2: Candidates

- Ravi, Mohan

Step 3: Apply Preferences

- Subject preference $\rightarrow$ Ravi $=$ He
- Object $\rightarrow$ Mohan $=$ him

Final Interpretation:

- Ravi greeted Mohan

Advanced Approaches

1. **Centering Theory**
   - Maintains discourse focus
   - Uses forward and backward centers
2. **Mitkov's Algorithm**
   - Knowledge-poor approach
   - Uses heuristics and indicators
3. **Machine Learning Approaches**
   - Neural networks for entity linking
   - Context-aware models (modern NLP)

Applications of Reference Resolution

1. **Chatbots**
   - Maintain conversational context
2. **Machine Translation**
   - Correct pronoun mapping across languages
3. **Question Answering**
   - Identify correct entity in text
4. **Information Extraction**
   - Link mentions of same entity
5. **Text Summarization**
   - Maintain clarity in summaries

Student Practice Problems

Resolve Reference:

- Sita met Geeta. She was happy.
- Ravi gave Mohan a book. He thanked him.

Find Ambiguity:

- Ravi met Mohan near his house.

Correct the Sentence:

- The car is new. She runs fast.

Explain Resolution:

Ravi saw a dog. It was barking loudly.

Advanced Questions (Exam-Oriented)

Theory Questions:

- Define reference resolution with examples
- Explain constraints and preferences

Analytical Questions:

- Resolve references in paragraph
- Identify ambiguous cases

10-Mark Questions:

- Explain reference resolution with algorithm
- Discuss approaches like centering and Mitkov

Constraints in Reference Resolution (Detailed)

Origin of Constraints:

- Derived from **linguistic agreement rules**
- Used in **anaphora and reference resolution**
- Act as **filters** to eliminate invalid candidates

What are Constraints?

- Constraints are **rules that restrict possible referents**
- They ensure that only **grammatically valid matches** are considered
- Used before applying preferences in resolution algorithms

Key Idea:

- Constraints answer: **“Which candidates are NOT possible?”**

Why Constraints are Needed

Without Constraints:

Ravi bought a printer. She is working.

- "She" could incorrectly refer to printer

With Constraints:

- Gender mismatch → invalid reference removed

Importance:

- Eliminates incorrect interpretations
- Reduces ambiguity
- Improves accuracy of NLP systems

In NLP Systems:

- First step in pronoun resolution
- Reduces search space for candidates

Types of Constraints

1. Person Agreement

- Pronoun must match grammatical person
- Example:
  - I, we $\rightarrow$ first person
  - you $\rightarrow$ second person
  - he, she, it $\rightarrow$ third person

2. Gender Agreement

- Pronoun must match gender of entity
- Example:
  - he $\rightarrow$ male
  - she $\rightarrow$ female
  - it $\rightarrow$ object/neutral

3. Number Agreement

- Singular vs plural consistency
- Example:
  - they $\rightarrow$ plural
  - it $\rightarrow$ singular

Examples with Explanation

Example 1: Gender Constraint

Ravi bought a laptop. She is working.

- “She” cannot refer to “laptop” → invalid

Example 2: Number Constraint

The students entered the class. He was late.

- “He” does not match plural “students”

Example 3: Person Constraint

Ravi said that I will come.

- “I” may not match Ravi (depends on context)

Worked Example (Step-by-Step)

Text:

Sita met Ravi. He was happy.

Step 1: Identify Pronoun

- "He"

Step 2: Possible Candidates

- Sita (female)
- Ravi (male)

Step 3: Apply Constraints

- Gender constraint → eliminates Sita

Final Result:

- "He" → Ravi

Constraints vs Preferences

Constraints:

- Hard rules (must be satisfied)
- Eliminate invalid candidates

Preferences:

- Soft rules (guidelines)
- Rank remaining candidates

Example:

Ravi met Mohan. He smiled.

- Constraints $\rightarrow$ both valid
- Preference $\rightarrow$ Ravi (subject)

Applications of Constraints

1. **Chatbots**
   - Prevent incorrect entity mapping
2. **Machine Translation**
   - Maintain correct gender/number agreement
3. **Question Answering**
   - Filter incorrect answers
4. **Information Extraction**
   - Link correct entities across text
5. **NLP Pipelines**
   - First stage before scoring candidates

Student Practice Problems

Identify Constraint Violation:

- The car is new. She runs fast.
- The boys are playing. He is happy.

Resolve Reference Using Constraints:

- Sita met Ravi. He smiled.

Correct the Sentence:

- The books are interesting. It is useful.

Explain Constraint Used:

Ravi saw a dog. It was barking.

Advanced Questions (Exam-Oriented)

Theory Questions:

- Define constraints in reference resolution
- Explain types of constraints with examples

Analytical Questions:

- Identify constraint violations in text
- Apply constraints to resolve references

10-Mark Questions:

- Explain constraints and their role in NLP
- Differentiate between constraints and preferences

Preferences in Reference Resolution (Detailed)

Origin of Preferences:

- Derived from **psycholinguistics and discourse analysis**
- Used in **anaphora resolution algorithms**
- Represent **heuristics (soft rules)** rather than strict constraints

What are Preferences?

- Preferences are **guidelines used to rank possible referents**
- Applied after constraints filter invalid candidates
- Help choose the **most likely antecedent**

Key Idea:

- Preferences answer: **"Which valid candidate is most likely?"**

Why Preferences are Needed

Without Preferences:

Ravi met Mohan. He smiled.

- Both Ravi and Mohan satisfy constraints

With Preferences:

- "He" $\rightarrow$ Ravi (subject preference)

Importance:

- Resolves ambiguity when multiple candidates exist
- Improves accuracy of reference resolution
- Mimics human interpretation behavior

In NLP Systems:

- Used in scoring mechanisms
- Helps rank candidate entities

Types of Preferences

1. **Recency Preference**
   - Most recently mentioned entity is preferred
   - Example:
     - Ravi met Mohan. Mohan smiled.
2. **Subject Preference**
   - Subject is more likely than object
   - Example:
     - Ravi met Mohan. He smiled. (He $\rightarrow$ Ravi)
3. **Parallelism Preference**
   - Similar grammatical roles are preferred
   - Example:
     - Ravi likes cricket. Mohan does too.

Examples with Explanation

Example 1: Subject Preference

Ravi met Mohan. He greeted him.

- "He" → Ravi (subject)
- "him" → Mohan (object)

Example 2: Recency

Ravi met Mohan near Sita. He was happy.

- "He" → Sita (most recent, but ambiguous)

Example 3: Parallelism

Ravi likes cricket. Mohan does too.

- "does too" → likes cricket

Worked Example (Step-by-Step)

Text:

Sita met Geeta. She was happy.

Step 1: Candidates

- Sita, Geeta

Step 2: Apply Constraints

- Both valid (female, singular)

Step 3: Apply Preferences

- Recency → Geeta
- Subject preference → Sita

Conclusion:

- Ambiguous → depends on context

Preferences vs Constraints

Constraints:

- Hard rules (must be satisfied)
- Eliminate invalid candidates

Preferences:

- Soft rules (heuristics)
- Rank valid candidates

Example:

Ravi met Mohan. He smiled.

- Constraints $\rightarrow$ both valid
- Preference $\rightarrow$ Ravi (subject)

Applications of Preferences

1. **Chatbots**
   - Resolve user references in conversation
2. **Machine Translation**
   - Select correct pronoun mapping
3. **Question Answering**
   - Choose correct entity from context
4. **Text Summarization**
   - Maintain clarity of references
5. **NLP Pipelines**
   - Used after constraints in ranking stage

Student Practice Problems

Resolve Using Preferences:

- Ravi met Mohan. He smiled.
- Sita met Geeta. She laughed.

Identify Preference Used:

- Ravi likes cricket. Mohan does too.

Find Ambiguity:

- Ravi met Mohan near his house.

Explain Reasoning:

Ravi gave Mohan a book. He thanked him.

Advanced Questions (Exam-Oriented)

Theory Questions:

- Define preferences in reference resolution
- Explain types of preferences with examples

Analytical Questions:

- Apply preferences to resolve ambiguity
- Compare different preference strategies

10-Mark Questions:

- Explain preferences with types and examples
- Differentiate between constraints and preferences

Pronoun Resolution Algorithm (Detailed)

Origin of the Algorithm:

- Developed in **Computational Linguistics and AI**
- Early work by **Hobbs (1978)** and later improved by researchers like **Mitkov**
- Based on combining:
  - Linguistic constraints
  - Heuristic preferences
  - Scoring mechanisms (salience)

What is Pronoun Resolution?

- Process of determining the **antecedent of a pronoun**
- Links pronouns (he, she, it) to correct entity
- Core task in **discourse processing**

Key Idea:

- Convert ambiguous pronouns into **clear references**

Why Pronoun Resolution is Important

Problem Without Resolution:

Ravi met Mohan. He was happy.

- "He" is ambiguous

Importance:

- Removes ambiguity in text
- Enables correct understanding of discourse
- Essential for machine understanding of language

Applications in NLP:

- Chatbots (conversation tracking)
- Machine Translation (correct pronoun mapping)
- Question Answering systems
- Text Summarization
- Information Extraction

Algorithm Overview

Steps of Pronoun Resolution:

1. Identify candidate antecedents
2. Apply constraints (filter invalid)
3. Assign scores (salience)
4. Select best candidate

Key Concept:

- **Salience Score** = importance weight of entity
- Higher score → more likely antecedent

Step 1: Identify Candidates

Definition:

- Find all possible entities that pronoun can refer to

Example:

Ravi met Mohan near Sita. He smiled.

Candidates:

- Ravi
- Mohan
- Sita

Explanation:

- All previous nouns are potential candidates
- No filtering yet

Step 2: Apply Constraints

Definition:

- Eliminate candidates that do not match grammar rules

Constraints Used:

- Gender agreement
- Number agreement
- Person agreement

Example:

Sita met Ravi. He smiled.

- Candidates: Sita, Ravi
- Gender constraint removes Sita
- Remaining: Ravi

Step 3: Assign Scores (Salience)

Definition:

- Assign importance score to each candidate

Factors Affecting Score:

- Subject $\rightarrow$ high score
- Recency $\rightarrow$ higher score
- Repetition $\rightarrow$ increases score
- Grammatical role

Example:

Ravi met Mohan. He smiled.

- Ravi $\rightarrow$ subject $\rightarrow$ higher score
- Mohan $\rightarrow$ object $\rightarrow$ lower score

Step 4: Select Best Candidate

Definition:

- Choose candidate with highest score

Example:

Ravi met Mohan. He smiled.

- Ravi $\rightarrow$ highest salience
- Final answer: “He” $=$ Ravi

Insight:

- Combines constraints $+$ preferences $+$ scoring

Worked Example (Very Detailed)

Text:

Suha bought a laptop. She liked it.

Step 1: Identify Pronouns

- She, it

Step 2: Candidates

- Suha, laptop

Step 3: Apply Constraints

- She $\rightarrow$ female $\rightarrow$ Suha
- it $\rightarrow$ object $\rightarrow$ laptop

Step 4: Assign Scores

- Suha $\rightarrow$ subject $\rightarrow$ high salience
- laptop $\rightarrow$ object $\rightarrow$ medium salience

Final Resolution:

- She $\rightarrow$ Suha
- it $\rightarrow$ laptop

Applications of the Algorithm

1. **Chatbots**
   - Track conversation entities
2. **Machine Translation**
   - Correct pronoun usage in target language
3. **Question Answering**
   - Identify correct answer entity
4. **Information Extraction**
   - Link mentions across text
5. **Text Summarization**
   - Maintain clarity of references

Student Practice Problems

Resolve Step-by-Step:

- Ravi met Mohan. He greeted him.
- Sita gave Geeta a book. She thanked her.

Find Ambiguity:

- Ravi met Mohan near his house.

Apply Algorithm:

The dog chased the cat. It ran away.

Advanced Questions (Exam-Oriented)

Theory Questions:

- Explain pronoun resolution algorithm
- Define salience with examples

Analytical Questions:

- Apply algorithm to resolve references
- Explain role of constraints and preferences

10-Mark Questions:

- Explain pronoun resolution algorithm with example
- Discuss salience scoring in detail

Centering Algorithm (Centering Theory) - Detailed

Origin of Centering Theory:

- Proposed by **Grosz, Joshi, and Weinstein (1995)**
- Developed in **discourse analysis and computational linguistics**
- Focuses on maintaining **local coherence in discourse**

What is Centering Theory?

- A framework to model how **attention shifts across sentences**
- Tracks the **main focus (center)** of discourse
- Helps resolve pronouns based on discourse focus

Key Idea:

- Discourse is easier to understand when **focus remains stable**

Why Centering is Important

Problem Without Centering:

Ravi met Mohan. Sita joined them. He smiled.

- "He" is ambiguous

With Centering:

- Focus helps identify most likely referent

Importance:

- Maintains discourse coherence
- Reduces ambiguity
- Models human understanding of text

In NLP Systems:

- Improves pronoun resolution
- Helps in dialogue systems
- Maintains context in conversations

Core Components of Centering

1. **Forward-Looking Center (Cf)**
   - List of entities in current sentence
   - Ranked by importance (subject ¿ object)
2. **Backward-Looking Center (Cb)**
   - The entity linking current sentence to previous one
   - Represents **current discourse focus**
3. **Preferred Center (Cp)**
   - Highest-ranked element in Cf
   - Usually the subject

Centering Transitions

Types of Transitions:

- **Continue:** Same focus maintained
- **Retain:** Focus remains but shifts slightly
- **Shift:** Focus changes completely

Preference Order:

- Continue $>$ Retain $>$ Shift

Insight:

- More stable transitions $\rightarrow$ better coherence

Working of Centering Algorithm

Steps:

1. Identify entities in each sentence (Cf)
2. Rank entities (subject highest)
3. Determine Cb (link with previous sentence)
4. Identify Cp (preferred center)
5. Determine transition type
6. Select referent based on highest-ranked center

Key Idea:

- Focus tracking helps resolve pronouns

Worked Example (Step-by-Step)

Text:

Ravi went to the market. He bought fruits. They were fresh.

Sentence 1:

- Cf = {Ravi, market}
- Cp = Ravi

Sentence 2:

- "He" → Ravi (Cb = Ravi)
- Cf = {Ravi, fruits}
- Transition = Continue

Sentence 3:

- "They" → fruits
- Cf = {fruits}
- Transition = Shift

Interpretation of Example

Observations:

- Sentence 1 → introduces Ravi
- Sentence 2 → maintains focus (Continue)
- Sentence 3 → shifts focus to fruits

Conclusion:

- Centering prefers **minimal shifts**
- Helps maintain smooth discourse flow

Key Insight:

- Pronouns tend to refer to **current center**

Applications of Centering Algorithm

1. **Pronoun Resolution**
   - Improves accuracy using discourse focus
2. **Dialogue Systems**
   - Maintains conversational context
3. **Text Summarization**
   - Preserves coherence in summaries
4. **Machine Translation**
   - Maintains consistent references
5. **Discourse Analysis**
   - Models flow of information

Student Practice Problems

Identify Centers:

- Ravi met Mohan. He greeted him.

Find Transition Type:

- Sita bought a book. She read it. It was interesting.

Resolve Using Centering:

- Ravi went home. Mohan followed him. He was tired.

Explain Focus Shift:

The dog chased the cat. It ran away.

Advanced Questions (Exam-Oriented)

Theory Questions:

- Explain centering theory with components
- Define Cf, Cb, and Cp

Analytical Questions:

- Identify transitions in discourse
- Apply centering to resolve pronouns

10-Mark Questions:

- Explain centering algorithm with example
- Discuss role of centering in discourse coherence

Mitkov's Algorithm (Detailed)

Origin of Mitkov's Algorithm:

- Proposed by **Ruslan Mitkov (1998)**
- Designed for **anaphora (pronoun) resolution**
- Known as a **knowledge-poor approach**

What is Mitkov's Algorithm?

- A method to resolve pronouns using **heuristics instead of deep linguistic knowledge**
- Does not rely on full syntactic or semantic parsing
- Uses **surface-level indicators** and scoring

Key Idea:

- Use simple rules to achieve **efficient and practical resolution**

Why Mitkov's Algorithm is Important

Problem with Traditional Methods:

- Require deep parsing
- Computationally expensive
- Language-dependent

Mitkov's Advantage:

- Fast and efficient
- Works with minimal linguistic knowledge
- Easily adaptable to different languages

Importance:

- Suitable for real-world NLP systems
- Works well in large-scale applications

Core Approach

Steps in Mitkov's Algorithm:

1. Identify candidate antecedents
2. Apply constraints (filter invalid)
3. Assign scores using indicators
4. Select candidate with highest score

Key Concept:

- Uses **heuristic scoring instead of deep analysis**

Indicators Used in Mitkov's Algorithm

1. **Repetition Indicator**
   - Frequently mentioned entity gets higher score
2. **Distance Indicator**
   - Closer entities preferred over distant ones
3. **Definiteness Indicator**
   - Definite noun phrases ("the") preferred over indefinite
4. **Grammatical Role Indicator**
   - Subject gets higher score than object
5. **Parallelism Indicator**
   - Similar structures preferred

Examples with Explanation

Example 1: Repetition

Ravi bought a laptop. Ravi uses it daily. It is fast.

- "Ravi" repeated → high score

Example 2: Distance

Ravi met Mohan near Sita. He smiled.

- Nearest entity gets preference

Example 3: Definiteness

I saw a dog. The dog was barking. It was loud.

- "the dog" gets higher importance

Worked Example (Step-by-Step)

Text:

Ravi met Mohan. He greeted him.

Step 1: Candidates

- Ravi, Mohan

Step 2: Apply Constraints

- Both valid (male, singular)

Step 3: Apply Indicators

- Subject (Ravi) → higher score
- Distance → both equal
- Repetition → none

Final Decision:

- "He" → Ravi
- "him" → Mohan

Comparison with Other Approaches

Mitkov's Algorithm:

- Knowledge-poor
- Uses heuristics
- Fast and efficient

Centering Theory:

- Focus-based
- Tracks discourse attention

Full NLP Parsing:

- Deep analysis
- More accurate but complex

Applications of Mitkov's Algorithm

1. **Chatbots**
   - Resolve references quickly
2. **Machine Translation**
   - Efficient pronoun resolution
3. **Information Extraction**
   - Link entities across text
4. **Large Text Processing**
   - Works efficiently on big datasets
5. **Real-Time Systems**
   - Low computational cost

Student Practice Problems

Apply Mitkov's Algorithm:

- Ravi met Mohan. He smiled.
- Sita gave Geeta a book. She thanked her.

Identify Indicator:

- The dog barked. The animal ran away.

Find Best Candidate:

- Ravi saw Mohan near Sita. He waved.

Explain Reasoning:

The boy saw a dog. The dog chased him.

Advanced Questions (Exam-Oriented)

Theory Questions:

- Explain Mitkov's algorithm
- What is knowledge-poor approach?

Analytical Questions:

- Apply Mitkov's algorithm step-by-step
- Compare with centering theory

10-Mark Questions:

- Explain Mitkov's algorithm with example
- Discuss indicators used in Mitkov's approach

Coherence

Definition:

- Logical connection between sentences

Difference:

- Cohesion = surface
- Coherence = meaning

Questions:

- Difference between cohesion and coherence?

Coherence Relations

- Explanation
- Cause
- Contrast
- Elaboration
- Parallel

Example: She was tired because she worked all day.
Questions:

- What are coherence relations?

Hobbs' Theory

Concept:

- Uses world knowledge

Example: He went to station. He boarded train.
Questions:

- Explain Hobbs' approach

Discourse Interpretation

Steps:

1. Logical form
2. Syntax
3. Knowledge encoding
4. Deduction
5. Interpretation
6. Best interpretation

Questions:

- Why multiple interpretations?

Local Coherence

Definition:

- Coherence between nearby sentences

Method:

- Abductive reasoning

Questions:

- What is local coherence?

Discourse Structure

Concept:

- Tree-like representation

Relations:

- Cause
- Contrast
- Elaboration

Questions:

- What is discourse structure?

Summary

- Discourse = beyond sentence
- Cohesion + Coherence
- Reference resolution is key
- Algorithms:
  - Centering
  - Mitkov

Exam Questions:

- Explain cohesion vs coherence
- Explain reference resolution
- Discuss Hobbs' theory